RT System: IoT & Electronics

1.System Summary

The **RT system** is a smart, fully automated blackwater recycling system that purifies domestic toilet blackwater into sterile, clear flushing water, creating a closed-loop sanitation cycle.

The system's control grid is divided into **10 Physical Tanks**, **18 Electronic Sensors (Inputs)**, and **17 Control Actuators (Outputs)**.

Sensors (18 Digital Inputs):

- 13 Liquid Level Float Switches / Gauges
- 6 Cabin Monitoring Sensors
- 2 Valve Position Feedback Switches

Actuators & Outputs (17 Digital Outputs):

- **11 Electric Fluid Pumps:** 3 lift pumps, 2 horizontal push pumps, 1 chemical inlet pump, 1 ECR circulation pump, 1 ECR discharge pump, 1 brine dosing pump, 1 UV circulation pump, and 1 distribution booster pump.
- **3 Aeration Blowers**: 1 biology aeration blower, 1 ECR gas-stripping blower, and 1 continuous advanced treatment blower.
- **3 Special Auxiliary/Treatment Units:** Electrolysis DC Power Supply, UV Lamp, and Motorized Discharge Valve.

System Console:

- **Visual Level Indication:** Displays intuitive high/medium/low visual colour code with level for the Septic Tank, Regulating Tank, Middle Water Tank , Advanced Treatment Tank, and ECR.
- **Error Indication:** Displays descriptive error names pinpointing the exact sensor or actuator experiencing a malfunction along with the precise nature of the failure.
- **Total Usage Metrics:** Counts the volume of total number of flushes in real-time.
- **Active Stage Indicator:** Displays the current path of the liquid and the actuators in operations (status).

2. System Architecture & Process Flow

The system is split into two physical domains: **The Toilet Cabins** (User Interaction) and **The Blackwater Treatment**

Below is the pathway and the associated IoT control devices interacting at each stage:

[TOILET CABINS]

- ├─► Biosensors & Key Switches (DS102, BS101)
- └─► Gravity Flow / Sewage Line

[BIOLOGICAL STAGE]

- ├─► 1. Septic/Collection Tank (T101)
  - └─► Septic Tank Lift Pump (B1011) ──► controlled by High/Low Floats (LS1011/12)

- ├─► 2. Regulating Tank (T102)
  - └─► Regulating Tank Lift Pump (B1021) ──► pumps to Treatment Biological treatment

- ├─► 3. Biological Treatment
  - ├─► Anaerobic Tank A & B (T104/T106) ──► Push Flow Pumps (B1041/61)
  - ├─► Aerobic Tank A & B (T105/T107) ──► Aeration Blower (AP1051)
  - └─► Sedimentation Tank (T108)

- └─► 4. Middle Water Tank (T109)
  - └─► Middle Water Tank Lift Pump (B1092) ──► pumps to Advanced Treatment

[CHEMICAL & STERILIZATION STAGE]

- ├─► 5. Advanced Treatment Tank (T111)
  - └─► Advanced Treatment Tank Blowers (AP1111-14) [Runs 24 Hours]

- ├─► 6. Electrochemical Reactor (ECR) System (T112)
  - ├─► ECR Water Inlet Pump (B1111)
  - ├─► Salt Dosing Tank (T113) & Salt Pump (B1131)
  - ├─► ECR Circulating Pump (B1121) & Blower (AP1121)
  - ├─► Electrode Power Supply (FP001)
  - └─► ECR Discharge Valve (EV1121) & Discharge Pump (B1122)

- └─► 7. Reuse Water Tank (T115)
  - ├─► UV Circulation Pump (B1151) & UV Lamp (UV1151)
  - └─► Reuse Water Booster Pump (B1152) ──► loops back to Toilet Flush (Closed Loop)

3. IoT Device, Sensor & Actuator (18)

To execute the control loops, the system relies on standard Input/Output (I/O) modules connected to a centralized **Programmable Logic Controller (PLC) / Edge Gateway** that communicates via Ethernet/Cellular to the Cloud.

2.1 Sensors (Inputs to the IoT System)

These devices monitor human interaction and physical tank liquid levels, providing **Digital Inputs (DI)** to the controller.

S.No.	Code	Device Description	Instrument Type	Set Values	Purpose
1	**BS101**	Toilet Biosensor	PIR / Infrared	On/Off	Detects if a human is in the cabin.
2	**DS101**	Door Lock Switch	Mech. Microswitch	On/Off	Detects when the door is locked from the inside.
3	**DS102**	User Button Switch	Auto-Reset Button	Momentary	Manually trigger lights/fans.
4	**LS1011**	Septic Tank High Level	Float Switch	90% Vol	Triggers Septic Tank Lift Pump B1011.
5	**LS1012**	Septic Tank Low Level	Float Switch	20% Vol	Dry-run protection for B1011.
6	**LS1021**	Regulating Tank High Level	Float Switch	90% Vol	Triggers Regulating Tank Lift Pump B1021
7	**LS1022**	Regulating Tank Low Level	Float Switch	20% Vol	Dry-run protection for B1021.
8	**LS1091**	Middle Tank High Level	Float Switch	90% Vol	Triggers Middle Water Tank Lift Pump B1092.
9	**LS1092**	Middle Tank Low Level	Float Switch	20% Vol	Dry-run protection for B1092.

10	**LS1111**	Adv. Treatment High Level	Side-mount Level Gauge	90% Vol	Inhibits upstream pump B1092.
11	**LS1112**	Adv. Treatment Low Level	Side-mount Level Gauge	20% Vol	Permits downstream ECR feed.
12	**LS1121**	ECR Electrode Box High	Side-mount Level Gauge	90% Vol	Ends the water inlet/dosing phase.
13	**LS1122**	ECR Electrode Box Low	Side-mount Level Gauge	20% Vol	Permits initial ECR water entry; stops ECR discharge.
14	**LS1131**	Salt Tank Low Level	Side-mount Level Gauge	20% Vol	Triggers alarm for manual salt replenishment.
15	**LS1151**	Reuse Water High Level	Float Switch	90% Vol	Triggers booster pump B1152 for reuse line.
16	**LS1152**	Reuse Water Low Level	Float Switch	20% Vol	Triggers backup safety mode (closes toilet cabins).
17	**EV1121-O**	Discharge Valve Open	Mech. Microswitch	On/Off	Confirms valve is fully open before pump starts.
18	**EV1121-C**	Discharge Valve Close	Mech. Microswitch	On/Off	Confirms ECR chamber isolation.

2.2 Actuators & Pumps (Outputs from the IoT System - 17)

These electrical hardware devices respond to commands from the controller as **Digital Outputs (DO)**.

S.No.	Tag	Equipment Name	Engineering Purpose

1	**B1011**	Septic Tank Lift Pump	Pumps raw sewage from collection to the Regulating tank.
2	**B1021**	Regulating Tank Lift Pump	Moves adjusted sewage into the aerobic/anaerobic train.
3	**B1041**	1# Push Flow Pump	Maintains horizontal movement through Anaerobic Tank A.
4	**B1061**	2# Push Flow Pump	Maintains horizontal movement through Anaerobic Tank B.
5	**B1092**	Middle Tank Lift Pump	Pushes liquid into the Advanced Treatment stage.
6	**B1111**	ECR Water Inlet Pump	Feed pump to transfer water into the chemical reactor (T112).
7	**B1121**	ECR Circulating Pump	Keeps water flowing over electrolysis electrodes during reactor run.
8	**B1122**	ECR Discharge Pump	Pushes electro-coagulated, clean water to the reuse tank.
9	**B1131**	ECR Salt Dosing Pump	Injects brine solution into ECR for electrical conductivity.
10	**B1151**	Reuse Tank Circ. Pump	Recirculates stored water through UV sterilizer.
11	**B1152**	Reuse Water Booster Pump	Pressurizes clean water lines to toilets for flushing.
12	**AP1051**	Anaerobic/Aerobic Blower	Controls aeration levels to maintain aerobic bacterial colonies.
13	**AP1121**	ECR Tank Blower	Aerates ECR water to keep particles suspended / strip gasses.
14	**AP1111**	Adv. Treatment Blower	Constant 24H blower to ensure advanced organic removal.
15	**EV1121**	Discharge Valve	Motorized valve that seals or drains the ECR chamber.

16	**FP001**	Electrode Power Supply	Activates DC power to titanium/iron electrode plates.
17	**UV1151**	UV Disinfection Lamp	High-intensity UV light kills any remaining pathogens.

4. Logical Interlocking & Automation Flows

To maintain unmanned system integrity, the PLC/Controller evaluates logical inputs and enforces strict operations. If an interlocking condition fails, the system immediately suspends dangerous tasks (e.g., pumping into a full tank, or running a pump dry).

US1: Cabin Lighting and Odor Exhaust Control

- **Inputs:** Door lock switch (DS101), Biosensor (BS101), Cabin button (DS102).
- **Outputs:** Cabin Light, Exhaust Fan (TF).
- **Logics:**
  - **Turn-On Condition:** If DS102 is pressed, **OR** a human is detected (BS101), **OR** the door is locked (DS101 is closed) $\rightarrow$ Turn **ON** lighting and TF.
  - **Turn-Off Condition:** If the door is unlocked (DS101 opens) **AND** biosensor (BS101) detects no presence $\rightarrow$ Wait for **30 seconds** (on-delay timer), then Turn **OFF** lighting and TF.

US2: Septic Tank Lift Control (T101 to T102)

- **Goal:** Pump waste from Collection/Septic tank (T101) to Regulating tank (T102).
- **Logics:**
  - **Start Pump (B1011 = 1):** If Septic Level is High (LS1011 = 1) **AND** Regulating Tank Level is NOT High (LS1021 = 0).
  - **Stop Pump (B1011 = 0):** If Septic Level reaches Low (LS1012 = 1) **OR** Regulating Tank Level becomes High (LS1021 = 1).

US3 & US4: Regulating Tank & Biological Train Aeration

- **Goal:** Pump sewage from Regulating Tank (T102) into the anaerobic/aerobic train while ensuring proper oxygen levels.
- **Logics:**
  - **Start Pump (B1021 = 1):** If the Regulating Tank is High (LS1021 = 1) **AND** Middle Water Tank is NOT High (LS1091 = 0).
  - **Aeration Blower (AP1051):**

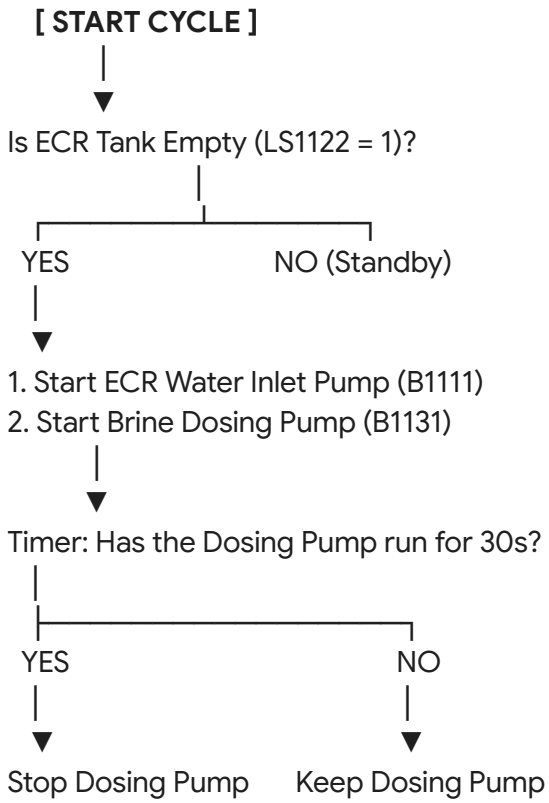
    - If B1021 is **ON** → AP1051 is forced **ON (100% duty cycle)**.
    - If B1021 is **OFF** → AP1051 runs **intermittently** (Runs for 20 mins, stops for 40 mins, repeats) to keep bacteria alive while saving energy.
  - **Push Flow Pumps (B1041, B1061):** Run intermittently on the opposite cycle when lift pump B1021 is off (Run 20 mins, stop 40 mins).

US5: Middle Water Tank Lift Control (T109 to T111)

- **Goal:** Prevent Middle Tank overflow and feed the Advanced Treatment stage.
- **Logics:**
  - **Start Pump (B1092 = 1):** If Middle Water Tank is High (LS1091 = 1) **AND** Advanced Treatment Tank is NOT High (LS1111 = 0).
  - **Stop Pump (B1092 = 0):** If Middle Water Tank reaches Low (LS1092 = 1) **OR** Advanced Treatment Tank is High (LS1111 = 1).

US6 & US7: ECR Electrolysis, Brine Dosing & Discharge Control

This is the most critical chemical stage. It ensures raw grey/black water undergoes electro-coagulation before reuse.



(B1131 = 0) Running (B1131 = 1)

Has ECR reached High Level (LS1121 = 1)?

YES NO

1. Stop Inlet Pump (B1111 = 0) Keep Filling
2. Start Circ. Pump (B1121)
3. Start Blower (AP1121)
4. Energize Electrodes (FP001)

Timer: Run Electrolysis for 30 mins

1. Stop Circ. Pump, Blower & Electrodes (B1121, AP1121, FP001 = 0)
2. Open ECR Discharge Valve (EV1121)
3. Confirm Valve Opened (EV1121-O = 1)
4. Start ECR Discharge Pump (B1122)

Has the ECR Chamber drained to Low Level (LS1122 = 1)?

YES NO

1. Stop Discharge Pump (B1122 = 0) Keep Draining
2. Close Discharge Valve (EV1121)

[END CYCLE]

US8 & US9: Reuse Water Treatment & Distribution

- **Goal:** Disinfect stored water and supply it back to toilets.
- **Logics:**
  - **Disinfection Loop (B1151 & UV1151):** Automatically triggered when the Low Level of

Reuse Water (LS1152) is detected (indicates water is present). They switch **OFF** if LS1152 vanishes (tank empty).

- **Toilet Shut-Off Interlock:** If Reuse Water Tank drops below empty threshold (LS1152 = 0) → Lock toilet cabin doors and change the LED indicator (LFD) to **"Out of Service"** to prevent flushing with no water.
- **Booster Pump (B1152):** Starts when High Level (LS1151 = 1) is detected to pressurize user flushing lines, and stops if water drops to Low Level.

5.Futuristic Options

Digitalisation

- Replace binary float switches with continuous ultrasonic/pressure level sensors for real-time % volume data instead of just High/Low triggers
- Digital flow meters on each pump line to log actual volume processed per cycle

Monitoring

- Real-time water quality sensors — pH, turbidity, TDS, ammonia — at key stages (post-biological, post-ECR, pre-reuse)
- Pump health monitoring — current draw, vibration, runtime hours — to predict failures before they happen
- Remote dashboard showing all 10 tanks, active actuators, and flush count live
- Alert system for salt tank low level (currently just an alarm), ECR electrode degradation, and UV lamp intensity drop

Control

- Adaptive ECR cycle timing based on actual water quality readings rather than fixed 30-minute timer
- Auto-adjust aeration blower duty cycle based on biological load

AI Integration

- Predictive maintenance model trained on pump runtime, current, and temperature data
- Anomaly detection to flag unusual patterns (e.g., ECR not reaching high level in expected time)
- Usage pattern learning to pre-emptively run treatment cycles before peak toilet usage hours
- AI-optimized brine dosing based on conductivity feedback instead of fixed 30-second timer

PLC to Microcontroller/Microprocessor Migration

- Replace the industrial PLC with an ESP32 (for smaller setups) or Raspberry Pi / Jetson Nano (for full processing power) — significantly lower cost and more flexibility
- Use dedicated I/O expander modules to handle all 18 digital inputs and 17 digital outputs since microcontrollers have limited GPIO
- Relay driver boards to safely switch pumps, blowers, and high-voltage actuators from 3.3V/5V logic
- Keep all existing interlocking logic (US1–US9) but implement it in firmware (C++/MicroPython) instead of ladder logic
- Watchdog timer in firmware to auto-restart the controller if it hangs — replicating PLC-grade reliability
- Modular architecture — one master microprocessor (Raspberry Pi) for logic + cloud, one or two ESP32 slave nodes for I/O handling via I2C/SPI

Cloud Integration

- Protocol: MQTT over WiFi/4G cellular SIM module (e.g., SIM7600) for reliable low-bandwidth real-time data publishing
- Cloud Platform: AWS IoT Core / Google Cloud IoT / Azure IoT Hub
- Data sent to cloud every cycle:
    - All 18 sensor states
    - All 17 actuator states
    - Tank levels (%, not just High/Low)
    - Flush count
    - ECR cycle status and duration
    - Pump runtime hours

Remote Monitoring

- Live tank level dashboard for all 10 tanks
- Active stage indicator — which cycle is currently running
- Historical trends — water processed per day, ECR cycles per week
- Alerts via push notification / SMS / email for faults, low salt, UV lamp warning
- Pump health logs — runtime hours, estimated service due dates

Remote Control

- Manually trigger or override any actuator (pump, blower, valve) from the app
- Force-start/stop ECR cycle remotely
- Adjust timer setpoints (ECR electrolysis duration, brine dosing time, aeration cycle)
- Put system in maintenance mode — safely disables all outputs for servicing
- Lock/unlock toilet cabins remotely